Kuha handoff

Branch **engine-integration**, as of 10 October 2026. For whoever picks up the next session (planned: Ollama and a sidebar navigation).

The backend now implements every endpoint in the AGENTS.md API contract, and all 66 backend tests pass. The frontend still uses stubs for example photos, XMP, finals and video clips, so those features work through the API but not yet in the UI. Read AGENTS.md and README.md first; this page only covers what they do not say.

1. State of the repository

Commit	What it holds
086c763	QA fixes: RAW capture time, honest export results, plain-language errors, no empty flash after organizing, duplicate React keys, input validation.
c42361b	Backend completion: example photos, video ingest with talking/scenery, XMP keywords, finals matching, highlights endpoint, and the example-photo scoring fix.

- Neither commit has been pushed unless the user did so after this document was written.
- **AGENTS.md is listed in .git/info/exclude**, so it is not tracked. The local copy has two extra lines in the API contract (GET /refs/{id}/image and GET /events/{id}/finals). Other machines will not have them.
- The data/ folder is gitignored and currently empty. There is no labelled test set on this Windows machine; results.md was generated on a Mac.

2. Running it

```
cd backend
.venv\Scripts\activate
python scripts/download_models.py (one time, needs internet)
uvicorn app.main:app --port 8000

cd frontend
npx vite --host 127.0.0.1 (see the port note below)

cd backend
python -m pytest -q (66 tests)
```

- **Port 5173 conflict.** Another project's dev server on this machine holds [::1]:5173, so *npm run dev* fails (strictPort). Binding Kuha to 127.0.0.1 works and is allowed by the backend CORS list.
- **FFmpeg and ExifTool** are installed and on the user PATH, but only terminals opened after the install see them. Without FFmpeg, videos are skipped with a log warning; without ExifTool, the XMP route returns 503 and finals matching keeps the ratings already stored. The tests that need them are skipped when they are missing.
- The frontend talks to http://127.0.0.1:8000 (frontend/src/api/client.ts).

3. Backend endpoints

Endpoint	Status	Notes
GET/POST /events, PATCH /events/{id}	Done	Blank names are rejected (422).
POST /events/{id}/template	Done	Replaces the event's shot items.
GET /events/{id}/shot_items	Done	Each item now carries refs: [ids].
POST /shot_items/{id}/refs	New	Multipart field 'files'. Stores a JPEG, embeds it, re-labels the whole event.
GET /refs/{id}/image	New	Serves a stored example photo.

Endpoint	Status	Notes
POST /events/{id}/ingest	Extended	Photos, RAW and now MP4/MOV. SSE stages: previews, embedding, videos, classifying, highlights.
GET /events/{id}/media, /highlights	Done	404 for a missing event. Videos carry roll and speech_ratio.
GET /events/{id}/coverage	Done	Missed-moment check.
GET /search	Done	A blank query returns an empty list.
POST /events/{id}/export	Done	Fails honestly: reports files that could not be copied; refuses a destination whose parent does not exist.
POST /events/{id}/xmp	New (Pro)	mode: lightroom or darktable. Backs up sidecars to .kuha_backup/, adds keyword tags only. Synchronous; returns {written}.
POST and GET /events/{id}/finals	New (Pro)	POST starts a job; GET returns matched, unmatched_finals, unedited_selects.

4. Decisions that differ from AGENTS.md

- **Example-photo scoring.** The spec formula blends raw cosines. Photo-to-photo cosines are far higher than photo-to-text ones, so a single example made its moment win every photo (this is the drop from 73.7% to 44.7% in results.md). ml/classifier.py now rescales each moment's example cosine to the mean and spread of its text cosine over the event's media before blending. Thresholds keep their meaning. The effect of examples is now modest: they settle close calls but do not override a strong text match. ALPHA_WITH_REFS still controls the weight.
- **Finals matcher.** Among the top 5 selects by cosine, the closest pHash wins if it is within 12 bits; otherwise the top cosine wins if it is at least 0.92. The literal rule could reject a heavily edited final.
- **Video capture time** comes from the container's creation_time via ffprobe, falling back to file mtime.

5. Known gaps and risks

- **Thresholds are untuned.** TAU_SCORE, TAU_MARGIN, TAU_COV and the new TAU_TALK (0.18) are guesses. results.md shows only 3 of 8 hidden moments flagged. The missed-moment check is the feature AGENTS.md says never to cut.
- **The scoring fix has not been measured.** Re-run scripts/eval.py on the machine that has the labelled set and compare the 'with example photos' line.
- **Talking clips are untested on real speech.** Only a generated clip with a tone was available; it came out as scenery, as expected.
- **scripts/eval.py** still prints 'not measured' for talking/scenery and finals accuracy. README results are all TBD.
- **RapidRAW** is offered in Settings but the XMP route accepts only lightroom and darktable.
- Example photos share the event's emb.npy with media; refs.emb_index and media.emb_index point into the same file.

6. Frontend work still open

All in frontend/src/api/data.ts unless noted. The typed API wrappers in client.ts already exist for most of these.

Stub or gap	What it needs
addExamples(eventId)	A file picker per moment, then api.addRefs(shotItemId, files) and a reload. MomentView.refs is always [] today; fill it from shot item refs via /refs/{id}/image. The ShotItem type lacks the refs field.
writeXmp(eventId, mode)	Call api.writeXmp and set xmpWritten. Decide what 'rapidraw' maps to.
rescanFinals(eventId)	A folder input on the Finals page, api.matchFinals, watchJob, then GET /events/{id}/finals (no wrapper yet) mapped into FinalsView.

Stub or gap	What it needs
clips: { A, B }	Hard-coded to 0. Fill from media where kind is 'video', grouped by roll.
pickFolder()	A browser cannot open a native folder picker. Either remove the buttons or add a backend route that opens one.
Add more photos later	In Personal mode the organize card disappears once an event has photos, so a second folder cannot be added from the UI.

7. Notes for the next session: Ollama and sidebar navigation

- **Guardrails still apply.** AGENTS.md forbids cloud AI and network calls at runtime. Ollama on localhost fits, but its model must be downloaded ahead of time (the equivalent of download_models.py) and the app must keep working with Wi-Fi off.
- **Fail soft.** If Ollama is not running, the feature should say so in plain language and the rest of the app should work.
- **Decide what Ollama is for before building.** Nothing in AGENTS.md or README.md defines a use for a language model yet. Add it to the README disclosure table, since the hackathon requires listing every model.
- **Plain language in Personal mode.** New labels go through frontend/src/lib/terms.ts. Error text shown to users comes from the backend's 'detail' string, so write those as full sentences.
- **Navigation today** is a top bar (Page in components/ui.tsx) with routes in App.tsx: /, /events/:id, /events/:id/export, /events/:id/finals, /settings. Mode lives in localStorage ('kuha.mode').
- The hackathon plan had a feature freeze at hour 9; check whether new features are still in scope.